

Optimizing Pharma Inventory using Machine Learning

- Pranav Ghildiyal

Objective

Scenario

Government run health centres need to maintain medicine stocks in limited space to cater to huge set of population. They need to maintain high availability of medicines so that they can immediately dispense it to the beneficiary.

Problem Statement

We need to create a model so we can optimize the inventory management ensuring high availability for high demand medicines. The constraints include an upper limit of amount of inventory given limited space, the lead time that vendor takes to deliver etc. Target is as low as possible Stock-Outs and Lost-Sales.

About the data set

- We are sourcing “Pharma Sales” data from [Kaggle](#) for this. The data is already clean and read to use.
- The data contains daily sales numbers across 6 years for medicines grouped into 8 groups. We can treat these groups as “Medicines” as a simplification for our business use case.
- We need Stock levels as well, which is missing in the available data. So, we have adding columns with Start of Day and End of Day stock numbers for all medicine (groups).
- We are assuming Ordering is done on Thursday, based on next 7 days demand. The delivery is added to Stock on Sunday morning, to simulate Vendor Lead Time.

Approach

- Get the source data
- Preprocessing - Add Stock levels to it
- Perform basic Exploratory Data Analysis to understand the data.
- Build initial models using Moving Average Baseline, SARIMA Optimization XGBoost, LightGBM in order to select suitable model.
- Proceed with the selected (best performing) model to reveal patterns based on different Vendor Lead Times, day of Ordering.
- Make inferences based on model.

Model Summary

Models Used

- XGBoost
- SARIMA Optimization
- LightGBM

Metrics Used (Performance Based)

- Total Stockout Incidents
- Total Lost Sales Units

Exploratory Data Analysis - I

--- Initial Exploratory Data Analysis ---

Total Number of Records: 2107

Total Number of Cols: 27

Medicine Groups : ['M01AB', 'M01AE', 'N02BA', 'N02BE', 'N05B', 'N05C', 'R03', 'R06']

First 10 rows of the dataset:

	Weekday	Year	Month	M01AB	M01AE	N02BA	N02BE	N05B	N05C	R03	...	N02BE_s_sod	N02BE_s_eod	N05B_s_sod	N05B_s_eod	N05C_s_sod
datum																
2014-01-01	Wednesday	2014	1	0.00	0.00	0.0	0.00	0.0	0.0	0.0	...	200.00	200.00	200.0	200.0	100.0
2014-01-02	Thursday	2014	1	0.00	3.67	3.4	32.40	7.0	0.0	0.0	...	200.00	167.60	200.0	193.0	100.0
2014-01-03	Friday	2014	1	8.00	4.00	4.4	50.60	16.0	0.0	20.0	...	167.60	117.00	193.0	177.0	100.0
2014-01-04	Saturday	2014	1	2.00	1.00	6.5	61.85	10.0	0.0	9.0	...	117.00	55.15	177.0	167.0	100.0
2014-01-05	Sunday	2014	1	4.00	3.00	7.0	41.10	8.0	0.0	3.0	...	394.15	353.05	167.0	159.0	100.0
2014-01-06	Monday	2014	1	5.00	1.00	4.5	21.70	16.0	2.0	6.0	...	353.05	331.35	159.0	143.0	100.0
2014-01-07	Tuesday	2014	1	0.00	0.00	0.0	0.00	0.0	0.0	0.0	...	331.35	331.35	143.0	143.0	98.0
2014-01-08	Wednesday	2014	1	5.33	3.00	10.5	26.40	19.0	1.0	10.0	...	331.35	304.95	143.0	124.0	98.0
2014-01-09	Thursday	2014	1	7.00	1.68	8.0	25.00	16.0	0.0	3.0	...	304.95	279.95	124.0	108.0	97.0
2014-01-10	Friday	2014	1	5.00	2.00	2.0	53.30	15.0	2.0	0.0	...	279.95	226.65	108.0	93.0	97.0

10 rows x 27 columns

Exploratory Data Analysis - II

--- Initial Exploratory Data Analysis Part-2 ---

Avg Monthly Sales M01AB :: 147.06162615741667

Avg Monthly Sales M01AE :: 113.8926200810139

Avg Monthly Sales N02BA :: 113.44734722223612

Avg Monthly Sales N02BE :: 875.0750376158334

Avg Monthly Sales N05B :: 258.5241319440417

Avg Monthly Sales N05C :: 16.859375

Avg Monthly Sales R03 :: 160.73365162044445

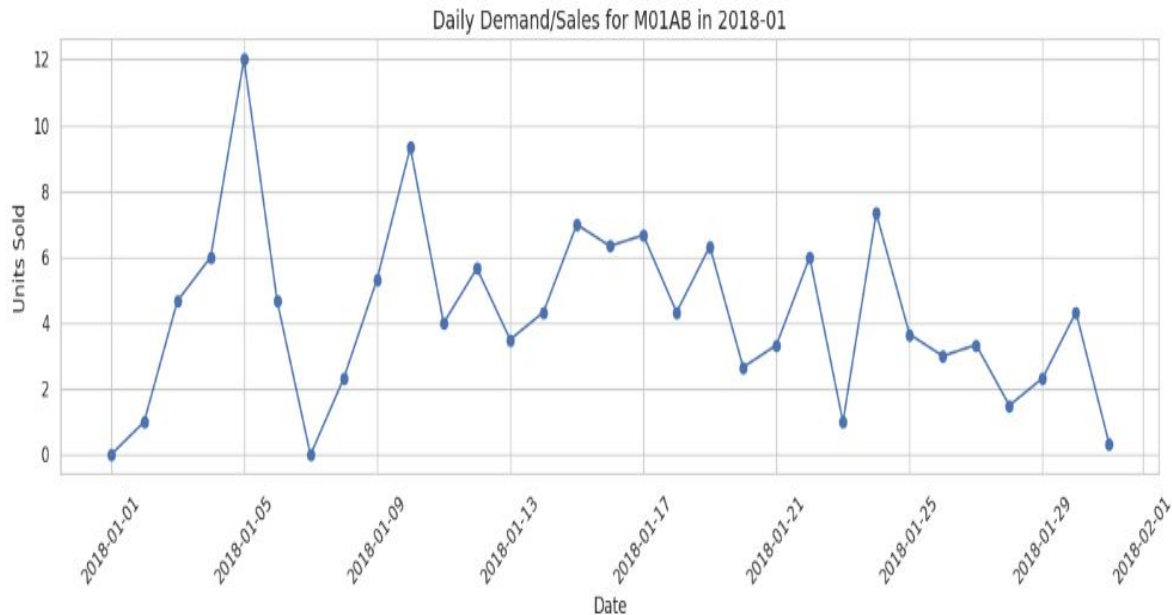
Avg Monthly Sales R06 :: 84.65024305555556

Highest Sale Month for M01AB

	Year	Month	M01AB
0	2014	11	138.080000
1	2015	10	195.810000
2	2016	8	211.130000
3	2017	1	181.902083
4	2018	11	172.290000
5	2019	7	181.000000

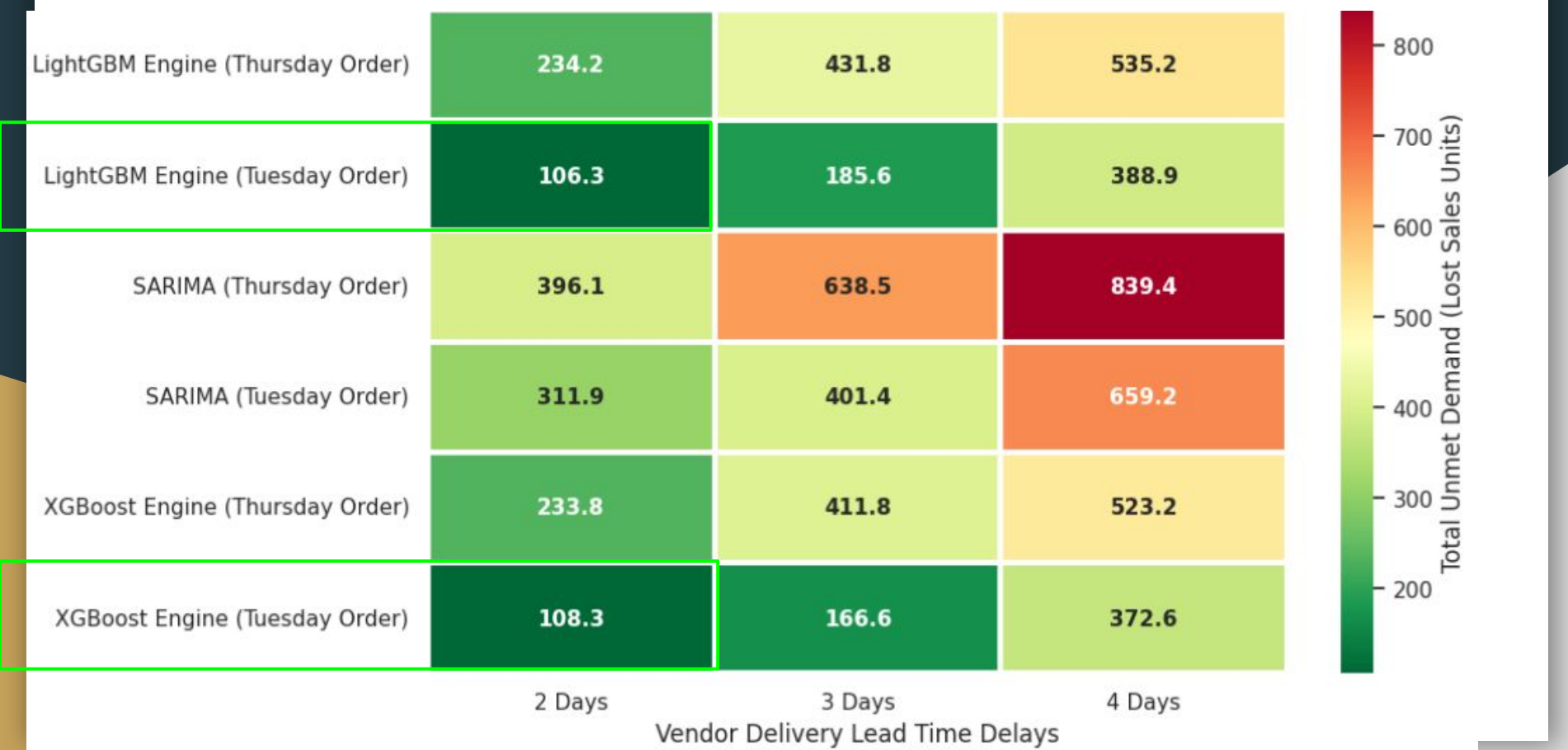
Monthly Sale for December for M01AB

	Year	M01AB
0	2014	131.90
1	2015	159.46
2	2016	169.32
3	2017	160.01
4	2018	147.71



Supply Chain Stress Matrix: Lost Sales across Models, Schedules & Delivery Lags

(Model Performance Analysis)



Doculte 8. Inference

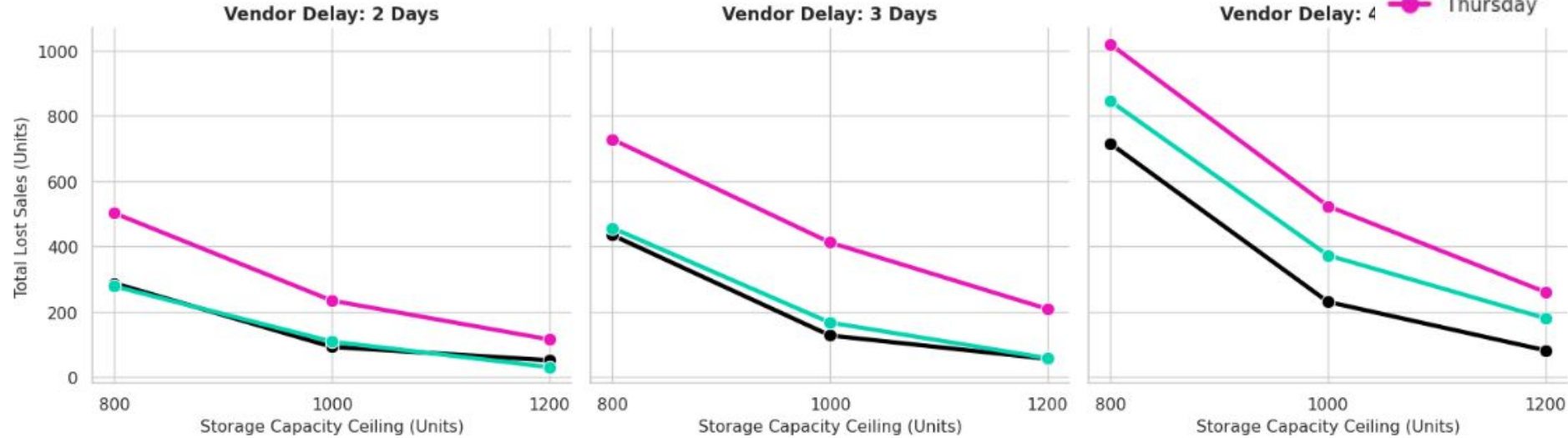
		Total Lost Sales (Units)			Total Stockout Incidents		
	Lead Time (Days)	2	3	4	2	3	4
Order Day	Storage Capacity						
Monday	800	286.96	434.24	714.67	34.00	50.00	76.00
	1000	92.13	127.27	230.15	13.00	19.00	30.00
	1200	51.13	55.65	81.64	6.00	8.00	9.00
Thursday	800	501.43	727.43	1,018.47	39.00	53.00	87.00
	1000	233.77	411.82	523.25	17.00	24.00	35.00
	1200	114.77	208.29	259.91	11.00	12.00	18.00
Tuesday	800	277.53	456.26	844.49	35.00	49.00	77.00
	1000	108.27	166.64	372.60	17.00	24.00	29.00
	1200	30.27	57.64	179.77	4.00	8.00	15.00

Tuesday is a more suitable day for Ordering Stock.

Vendor delivery lead time and Stockouts are directly proportional.

Storage capacity and Stockouts are inversely proportional.

Results & Inference (Impact on Lost Units)

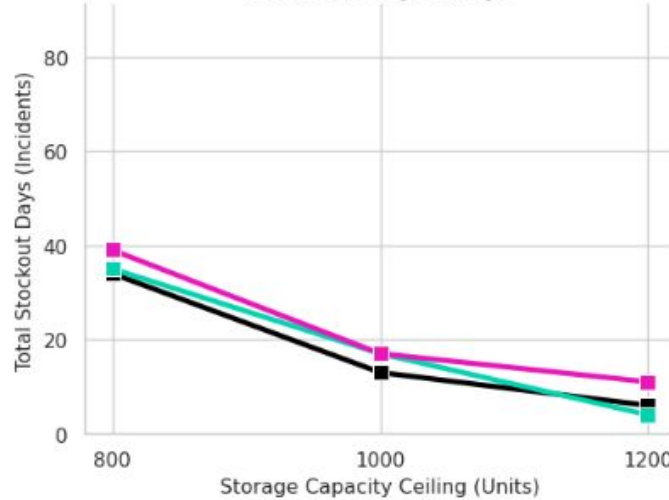


Vendor delivery lead time and Stockouts are directly proportional.
Storage capacity increase can compensate well.

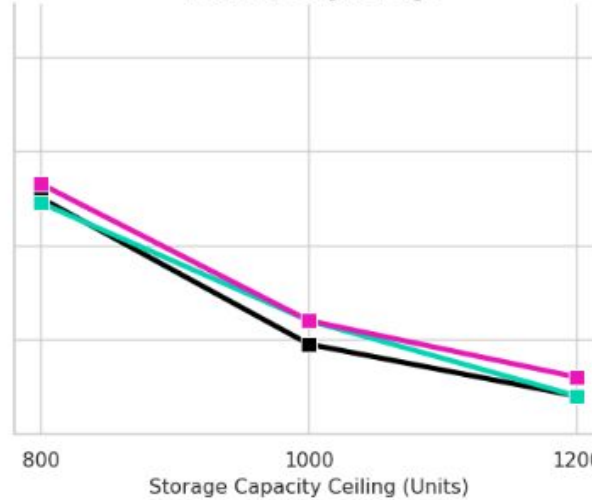
Results & Inference (Days having StockOuts)

Order Day
● Monday
● Tuesday
● Thursday

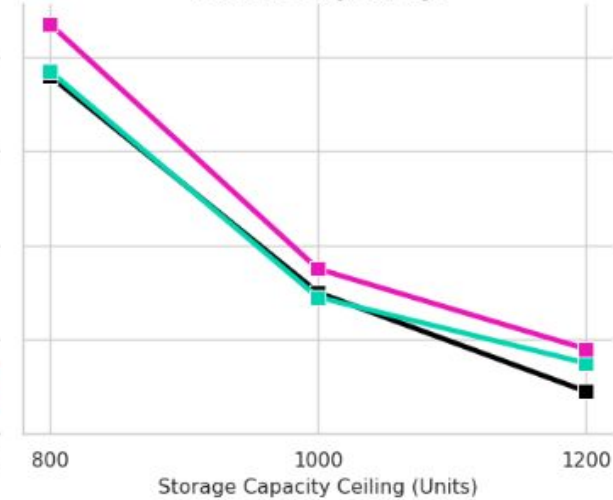
Vendor Delay: 2 Days



Vendor Delay: 3 Days



Vendor Delay: 4 Days



Vendor delivery lead time and Stockouts are directly proportional.
Storage capacity increase can compensate well.

References and Next Steps

Links

Kaggle : [initial-source-dataset](#)

Github : [Pharma-Shop-Inventory-Optimization-git-hub](#)

LightGBM: A Highly Efficient Gradient Boosting Decision Tree

XGBoost Optimization: A Scalable, End-to-End Tree Boosting System for Efficiency and Performance

SARIMA Optimization: Advanced Statistical Forecasting for Complex Seasonal and Trend-Driven Time Series.

Next Steps

Multi-day ordering – We can explore multi-day ordering and compare its performance with current once weekly approach.

Factor in stockouts/missing medicines – These health center pharmacies usually order the missing medicines which are stocked for a specific period (7days) for the beneficiary to collect anytime once received.